

Pulmonary Ventilation and Perfusion ($\dot{V}/\dot{Q}$)

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 Adapted from: [Pulmonary Ventilation and Perfusion](#)

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Learning Objectives (5)

Versions

After completing this brick, you will be able to:

- 1 Describe the processes of ventilation and perfusion.
- 2 Explain regional differences in pulmonary ventilation and perfusion in the upright healthy lung.
- 3 Explain the ventilation/perfusion ($\dot{V}/\dot{Q}$) ratio and its physiological significance.
- 4 Identify the lungs' West zones and the $\dot{V}/\dot{Q}$ ratio of each.
- 5 Compare shunt and dead space in healthy vs pathologic states.

**CASE CONNECTION**

CR is a 50-year-old man who presents to the emergency department with sudden-onset shortness of breath and pleuritic chest pain. His respiratory rate is 24 breaths/min, and his oxygen saturation is 87% on room air. Lung exam is clear and the chest X-ray is normal. Given the acute dyspnea, pleuritic pain, hypoxemia, and unremarkable lung exam, you suspect pulmonary embolism and initiate supplemental oxygen via nasal cannula while ordering a CT pulmonary angiogram (CTPA) for definitive evaluation.

If your diagnosis is correct, do you expect the patient's hypoxia to respond to supplemental O_2 ? Are there hypoxic conditions that do not respond well to supplemental O_2 ?

GO TO CONCLUSION ↓

What Are the Processes of Pulmonary Ventilation and Perfusion (LO1)?

The lungs' primary role is gas exchange: delivering oxygen to the blood and removing carbon dioxide. This depends on two fundamental processes:

- **Ventilation** ($\dot{V}$) is the movement of air into and out of the alveoli, and depends on the mechanics of breathing, including diaphragm and intercostal muscle contraction. Air moves due to pressure gradients between the atmosphere and alveoli, created by expansion and recoil of the lungs and chest wall. At rest, alveolar ventilation is ~4 L/min. The efficiency of ventilation depends on airway patency (openness), respiratory muscle function, and lung compliance. Ventilation is readily

assessed using expired CO_2 levels (the alveolar ventilation equation is discussed elsewhere).

- **Perfusion** ($\dot{Q}$) is the delivery of deoxygenated blood to alveolar capillaries for subsequent oxygenation, and in the lungs refers to the flow of blood from the right heart into the pulmonary capillaries. Perfusion is influenced by gravity, resulting in greater blood flow to lower lung regions. Additionally, local factors such as alveolar oxygen tension affects regional perfusion through hypoxic pulmonary vasoconstriction, which redirects blood flow away from poorly ventilated areas. Recall that the entire cardiac output (~5 L/min at rest) passes through the lungs, and that the pulmonary circulation is a low-pressure system that allows for efficient gas exchange across the thin, fragile alveolar-capillary membranes.

Together, the ventilation and perfusion enable oxygen to diffuse from alveoli into blood and carbon dioxide to diffuse out. Gas exchange depends not just on the presence of air or blood—but on their matched flows. If one is impaired, oxygen delivery to tissues is compromised, even if the other is functioning normally.

Perfusion refers to the flow of blood through the alveolar capillaries.
Ventilation involves the flow of air into and out of the alveoli.

How Are Ventilation and Perfusion Distributed in a Healthy Lung (LO2)?

In the upright position, both ventilation and perfusion increase from the lung apex to the base, primarily due to the "pulling" effects of gravity. These gravitational effects create predictable regional variations in $\dot{V}/\dot{Q}$ ratios throughout the healthy lung (Figure 1). Regional **perfusion** increases more dramatically than ventilation (Figure 1), with gravity creating a pressure gradient that directs blood flow to the lower lung regions.

Ventilation also increases toward the lung base, though less dramatically than perfusion (Figure 1). The gravitational effect on the pleural fluid allows alveoli in the apex to expand while compressing those at the base. These compressed basal alveoli have smaller resting air volumes (size) but higher compliance (expand more), allowing greater expansion with air during inspiration.

ventilation.

smaller alveoli that are more compliant, driving an increase in fluid, causing higher intrapleural pressure at the base resulting in of gravitational effects on the blood. Gravity also affects the pleural Perfusion increases at the base of the lungs when standing because

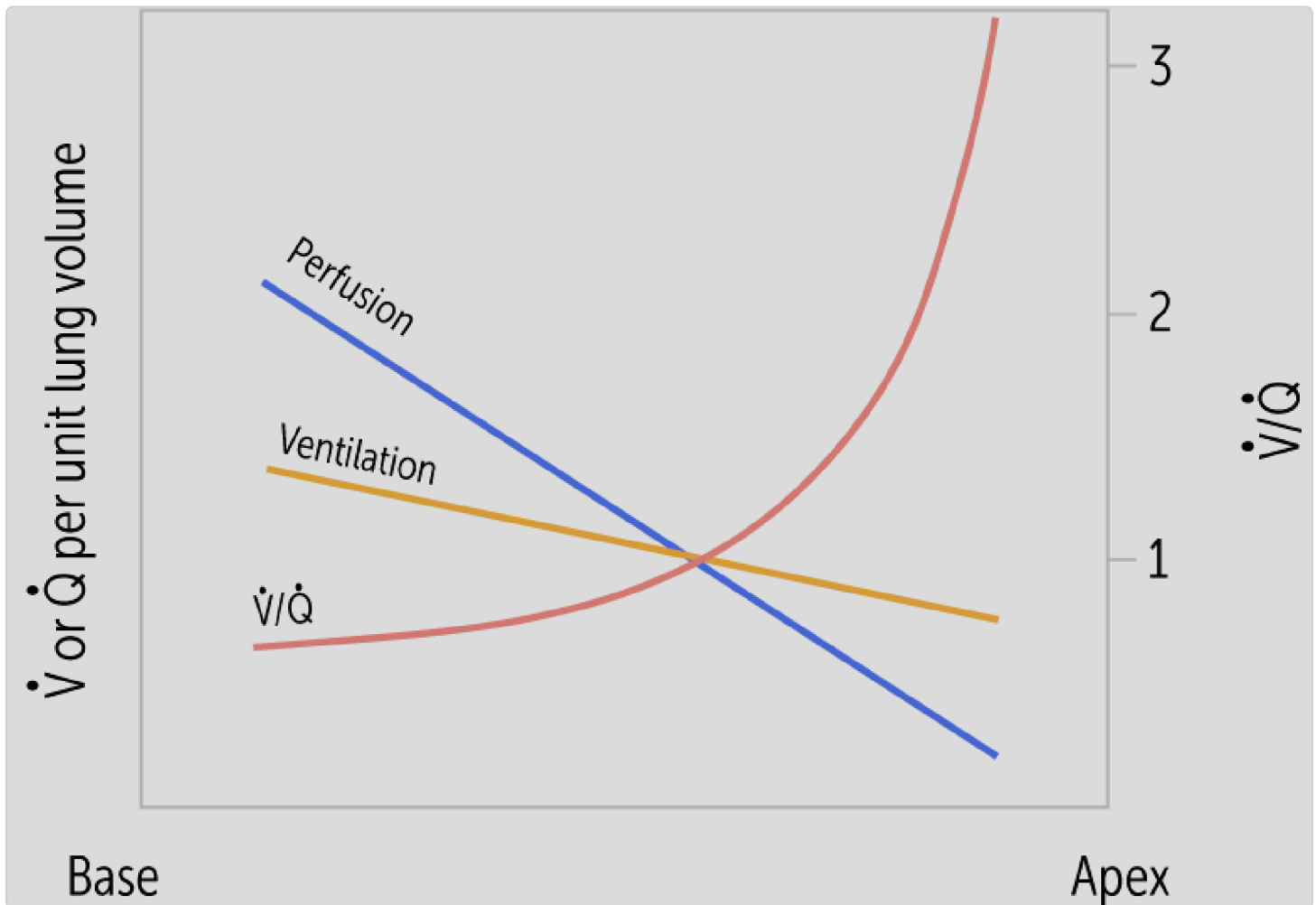


Figure 1 Ventilation and perfusion, and the $\dot{V}/\dot{Q}$ ratio in the upright lung

What Is the Ventilation/Perfusion ($\dot{V}/\dot{Q}$) Ratio (LO3)?

The **ventilation/perfusion ($\dot{V}/\dot{Q}$) ratio** compares alveolar airflow to blood flow, indicating how well ventilation and perfusion are matched in specific lung regions. Since both ventilation and perfusion increase at the base and decrease at the apex, they remain fairly well matched throughout the healthy

lung. This matching is critical for efficient gas exchange, where oxygen can effectively diffuse from alveolar air into capillary blood while carbon dioxide moves in the opposite direction. However, impairment of either ventilation or perfusion compromises oxygen delivery to tissues, even if the other is functioning normally.

Normal $\dot{V}/\dot{Q}$: The normal $\dot{V}/\dot{Q}$ ratio is approximately 0.8 (alveolar ventilation ~4 L/min, perfusion ~5 L/min), indicating well-matched ventilation and perfusion. The $\dot{V}/\dot{Q}$ ratio of the lung has a reverse "J" shape going from the base to the apex in an upright lung with the highest ratios at the apex and the lowest ratios at the base (Figure 1).

- The **apex of the lung** demonstrates high $\dot{V}/\dot{Q}$ ratios (>3.0) because ventilation is relatively preserved while perfusion is minimal due to low hydrostatic pressure (Figure 1).
- The **middle regions** of the lung approach the ideal $\dot{V}/\dot{Q}$ ratio of approximately 0.8, providing optimal gas exchange efficiency.
- The **lung base** shows low $\dot{V}/\dot{Q}$ ratios (<1.0) because perfusion greatly exceeds ventilation, leading to slightly incomplete oxygenation of blood in these regions (Figure 1). Despite this apparent inefficiency, the lung base contributes significantly to overall gas exchange due to its large volume and high blood flow.

CLINICAL CORRELATION

The high $\dot{V}/\dot{Q}$ at the apex of the lungs creates an oxygen-rich environment that favors the growth of aerobic organisms, explaining why tuberculosis reactivation typically occurs in the upper lobes.

are matched in a lung region.

The $\dot{V}/\dot{Q}$ ratio is a comparison of how well ventilation and perfusion

What Are West Zones (LO4)?

The West zones were discussed in the [Pulmonary Circulation](#) brick, and here we relate them to **ventilation and perfusion**. The cross-sectional diagrams on the right show the pulmonary vasculature (arteries in red, veins in blue) and alveoli, illustrating how blood flow changes across zones ([Figure 2](#)).

Alveoli are more expanded at the apex than the base, while perfusion increases from apex to base, as shown by the progressively larger vessels.

- **Zone 1: Apex**- This region does not typically occur in healthy lungs. In this region, the alveoli are expanded and cannot expand much more and have reduced ventilation. This, plus the region's lower hydrostatic pressure, means that perfusion is relatively low, and the $\dot{V}/\dot{Q}$ ratio is >1 . This relative hyperventilation means that the blood leaving this zone has the highest partial pressure of arterial oxygen (PaO_2) and the lowest partial pressure of arterial carbon dioxide (PaCO_2).

- **Zone 2: Middle-** Here the $\dot{Q}$ increases to match $\dot{V}$, and as a result, the $\dot{V}/\dot{Q}$ ratio in zone 2 is 0.8. This means that almost all the inspired air and blood flow in this zone participate in gas exchange.
- **Zone 3: Base-** At the lung's base, both P_V and P_a increase due to gravity, and they both exceed alveolar pressures (P_A). The $\dot{V}/\dot{Q}$ ratio typically is <1 in this zone, which leads to slightly limited ventilation in relation to perfusion, resulting in the blood leaving this zone having the lowest PaO_2 and the highest partial PaCO_2 .

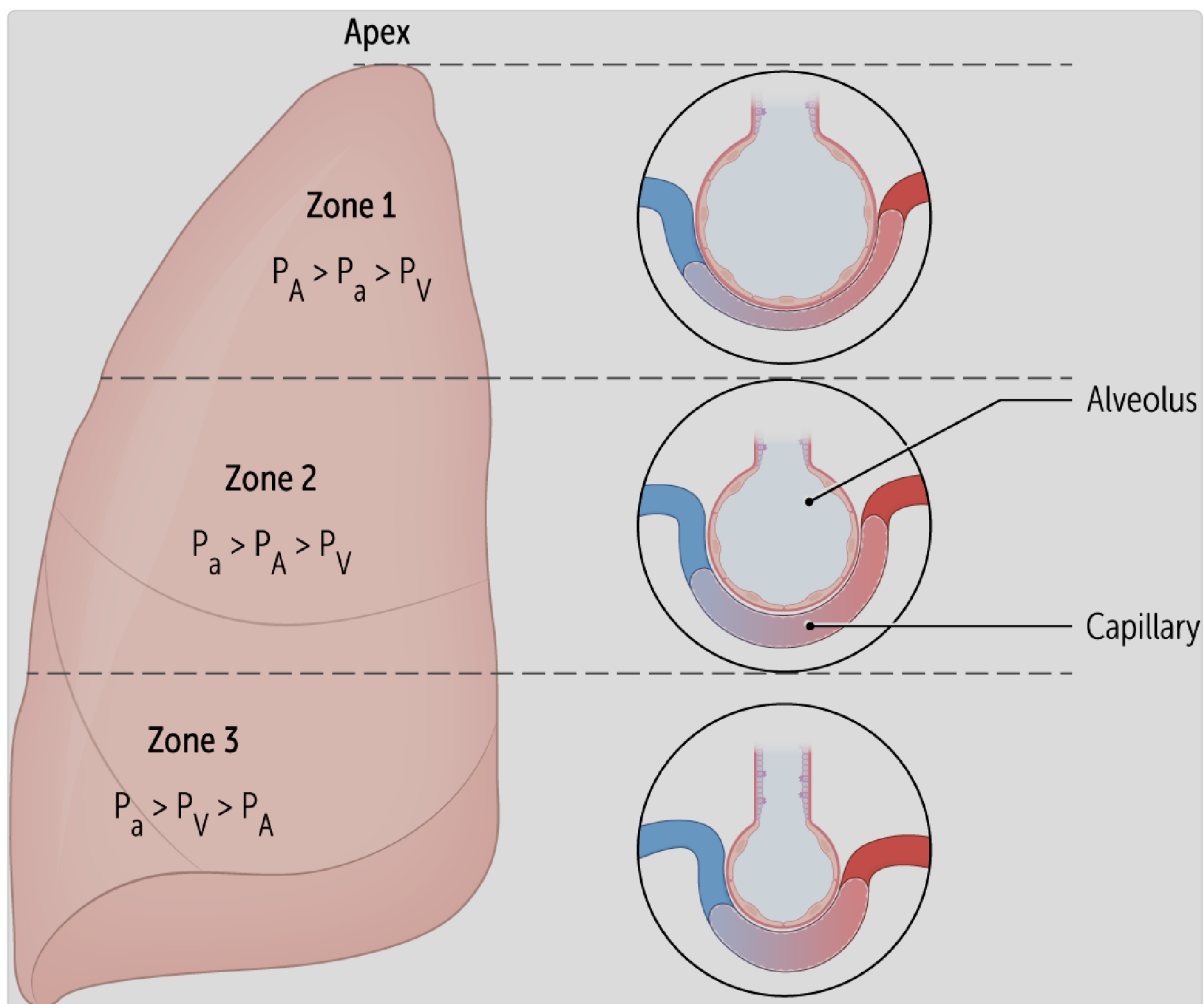


Figure 2. West lung zones

The net result is that the entire lung produces normal CO_2 and O_2 levels. This is because West zones with suboptimal $\dot{V}/\dot{Q}$ ratios are matched by zones with $\dot{V}/\dot{Q}$ ratios around 0.8. However, in disease states in which one lung region may be obstructed, not perfused with blood, or even resected (e.g., after surgery), this balance can be disrupted.



CLINICAL CORRELATION

West Zone 1 mainly exists when perfusion falls or alveolar pressure rises. In **hemorrhagic shock**, low circulating volume reduces capillary pressure, converting more alveoli into Zone 1 and increasing dead space. **Positive-pressure ventilation** can do similar, but by elevating alveolar pressure enough to collapse capillaries. In both cases, blood flow is displaced from the alveoli, driving $\dot{V}/\dot{Q}$ toward infinity and impairing gas exchange.

Ventilation exceeds blood flow in West zone 1, at the Apex of the lung.

Shunt and Dead Space and $\dot{V}/\dot{Q}$ mismatch (LO5)

$\dot{V}/\dot{Q}$ **mismatch** occurs when this ratio deviates significantly from normal, impairing oxygenation and CO_2 elimination. **Shunt** and **dead space** represent extreme forms of $\dot{V}/\dot{Q}$ mismatch that have distinct physiological characteristics and clinical implications (**Figure 3**). Shunt and dead space (D) are therefore the two extremes of the $\dot{V}/\dot{Q}$ ratio continuum.

- **Shunt** ($\dot{V}/\dot{Q} \approx 0$): Blood flow (perfusion) without ventilation of alveoli
- **Dead space** ($\dot{V}/\dot{Q} \approx \infty$): Ventilation of alveoli without perfusion

the respiratory system without ventilation.

ventilated without perfusion, while shunts are well-perfused areas of
Dead space describes areas of the respiratory system that are

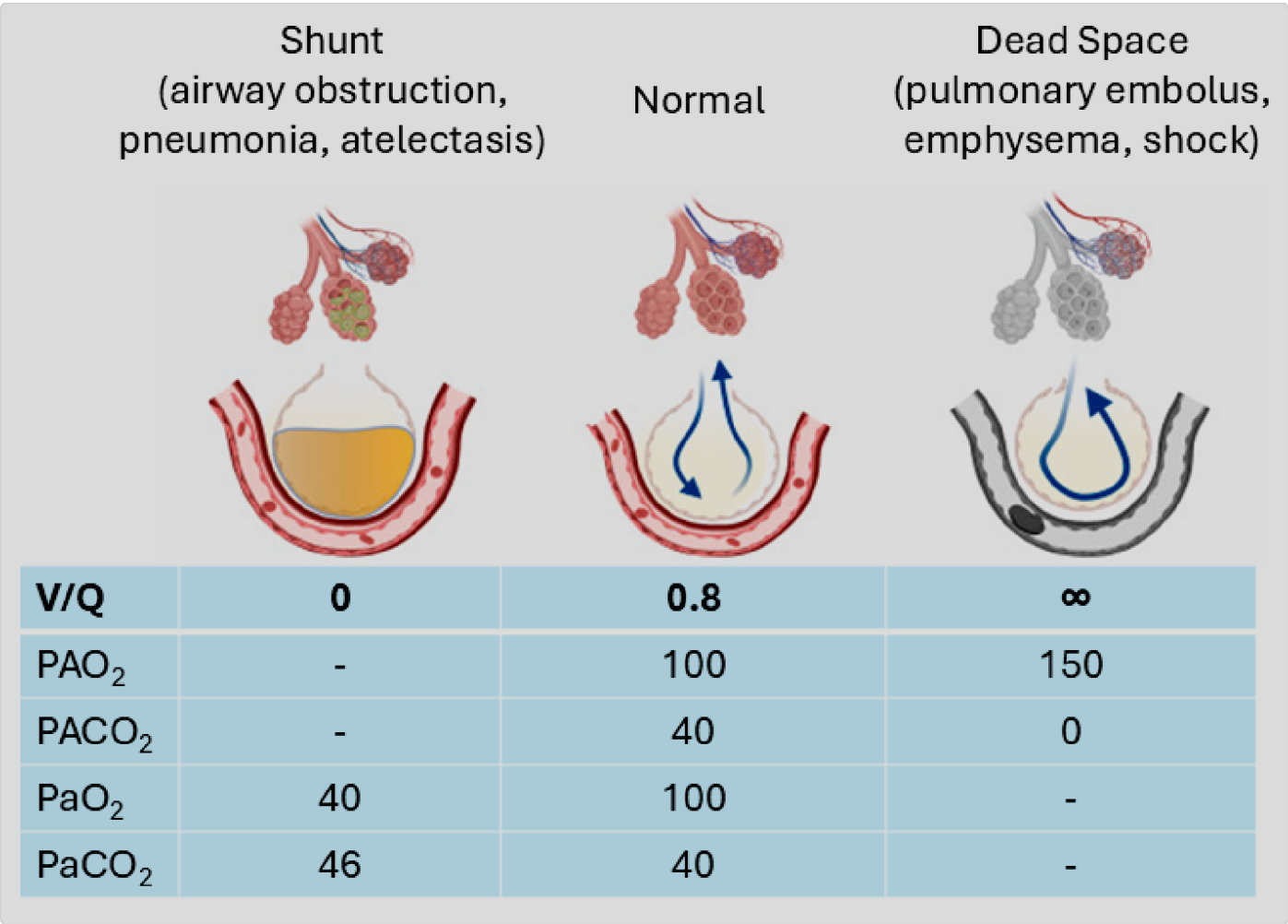


Figure 3. Effect of $\dot{V}/\dot{Q}$ mismatch on alveolar gas values

Shunt

Shunt occurs when blood flows through the pulmonary circulation without participating in gas exchange, resulting in a $\dot{V}/\dot{Q}$ ratio approaching zero (Figure 3). In shunt regions, alveoli may be collapsed, filled with fluid, or completely obstructed, preventing air from reaching the gas exchange surface. Blood leaving these regions maintains its venous characteristics, low oxygen content (approximately 40 mmHg) and high carbon dioxide content (46 mmHg).

Physiological shunt exists normally in healthy individuals, accounting for 2-5% of cardiac output. This small amount of normal shunting occurs

because the lungs have dual blood supplies: the bronchial circulation from the aorta (systemic circulation) and the pulmonary circulation from the right ventricle (**Figure 4**). Bronchial arteries branch off the systemic circulation (the aorta) and supply oxygenated blood to the conducting airways (bronchi and bronchioles) and other non-alveolar tissues of the lung. Much of this shunted blood then enters the pulmonary veins and returns to the left side of the heart without oxygenation.

Clinical Effects of Shunt: A shunt produces hypoxemia when poorly oxygenated blood from non-ventilated regions mixes with well-oxygenated blood from normal lung areas. While small amounts of shunting occur normally, large shunts cause arterial hypoxemia that cannot be corrected by increasing ventilation, to normal lung areas due to the nonlinear oxygen-hemoglobin dissociation curve. Shunting is characterized by **poor response to supplemental oxygen** because the shunted blood never contacts ventilated alveoli.

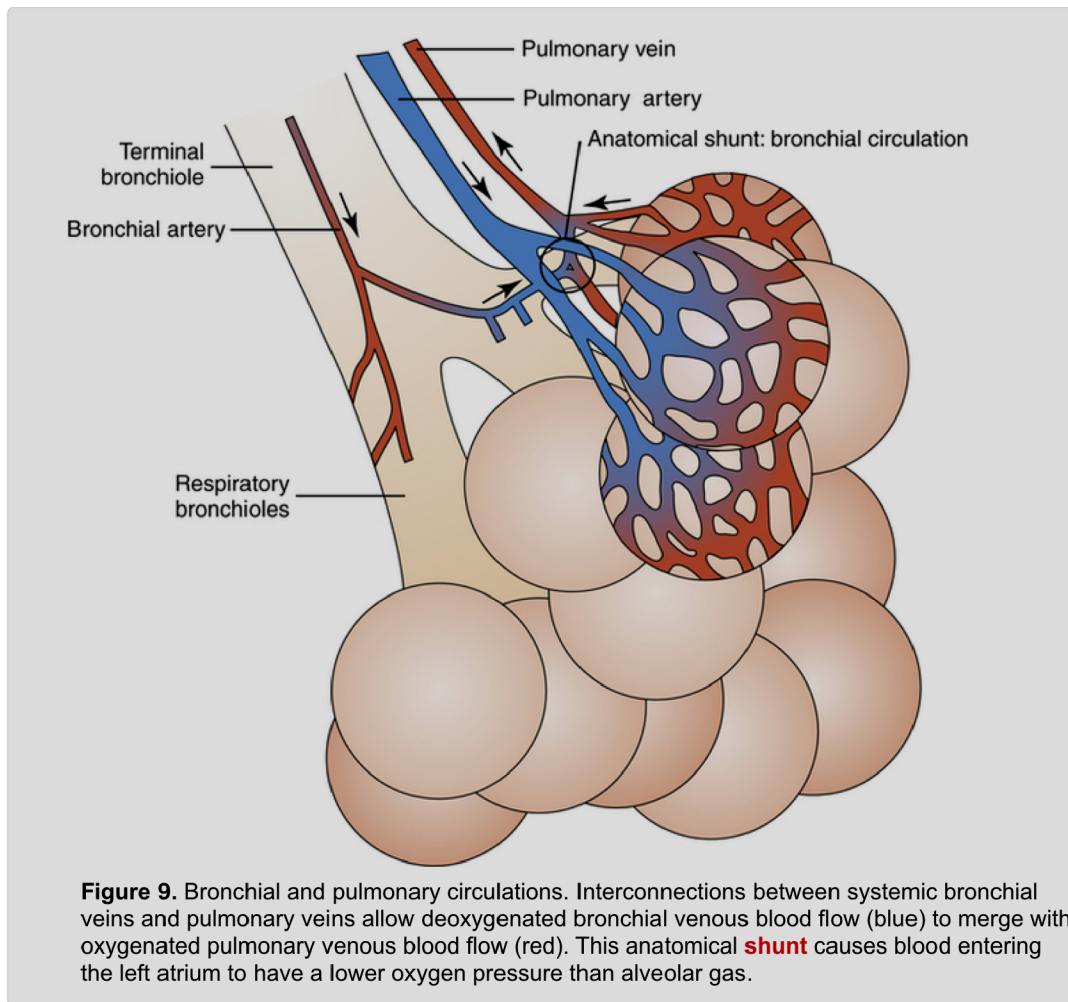


Figure 4. Bronchial vs alveolar blood supply

Pathological shunt develops in conditions such as congenital heart disease with right-to-left shunting, or disease states due to the common causes below:

- **Mucus plugging**, which can obstruct the alveolus, preventing ventilation. Examples are pneumonia, and cystic fibrosis.
- **Alveolar fluid** from cardiogenic pulmonary edema, acute respiratory distress syndrome, heart disease, or pneumonia, which can fill the alveolus, blocking ventilation.
- **Lung collapse** (atelectasis), which can collapse the alveoli, as seen in pneumothorax and other pulmonary diseases.



CLINICAL REASONING

Did You Know? With ventilation, not all inspired air reaches alveoli. Some remains in conducting airways—this is anatomical dead space (~150 mL).

Clinical Insight: Conditions such as pulmonary embolism or low cardiac output decrease perfusion, impairing gas exchange even if ventilation is normal.

Dead space

Dead space is the opposite of shunting. The effects of dead space on $P_A\text{CO}_2$ and expired CO_2 ($P_E\text{CO}_2$) and was discussed in the [Alveolar ventilation and dead space brick](#). Here the dead space is discussed in relation to ventilation and perfusion, and it represents areas of the lungs that are ventilated but not perfused, leading to a $\dot{V}/\dot{Q}$ ratio near infinity (∞) ([Figure 3](#)). In dead space regions, alveolar gas composition resembles inspired air, with oxygen content of approximately 150 mmHg and minimal carbon dioxide ($P_A\text{CO}_2 = 0$ mmHg). This occurs because no blood flow is available to extract oxygen or deliver carbon dioxide for elimination.

Anatomical dead space includes the conducting airways (trachea, bronchi, bronchioles) that transport air but do not participate in gas exchange, and typically totals approximately 150 mL in healthy adults.

Alveolar dead space (also called pathological dead space) primary occurs during diseases where alveoli are ventilated but not perfused. All of these reduce blood flow to the lung (and can increase the apical West zone 1).

Clinical effects of dead space, where there are alveoli with no perfusion, results in no elimination of CO_2 . While the arteriolar blood is not shown for those alveoli, if the dead space is large enough, there is a global reduction in CO_2 exhalation and elimination ($P_{\text{E}}\text{CO}_2$). Hypercapnia develops when $\dot{V}/\dot{Q}$ mismatch is severe enough to impair overall carbon dioxide elimination. However, there is ventilatory compensation. Unlike oxygen, carbon dioxide transport follows a more linear relationship, allowing increased ventilation to normal lung regions to partially compensate for impaired CO_2 elimination from mismatched areas. This explains why hypercapnia typically occurs later in the course of $\dot{V}/\dot{Q}$ mismatch diseases and indicates more advanced pathophysiology.



CLINICAL CORRELATION

Alveolar dead space can increase in disease states such as pulmonary embolism, emphysema (destruction of alveolar capillaries), heart failure, or hemorrhage (low cardiac output or volume). This significantly increases the work of breathing and impairs carbon dioxide elimination. Patients with high dead space fractions require increased minute ventilation to maintain normal arterial carbon dioxide levels. Dead space may improve with interventions that restore perfusion, such as anticoagulation for pulmonary embolism or hemodynamic support for shock.

$\dot{V}/\dot{Q}$ mismatch and deadspace are responsive to supplemental oxygen. An increase in FiO_2 will increase the alveolar O_2 in the regions that receive additional blood flow diverted from the deadspace areas. The affected alveoli with mismatched $\dot{V}/\dot{Q}$ still receive some amount of ventilation which can be

leveraged by an increased inspired oxygen (FiO_2) to increase oxygen transfer into the blood.

that is diverted from the unperturbed regions.

FiO_2 in the healthy lungs can better tolerate the increased blood flow some ventilation with supplemental O_2 . In deadspace, the additional underventilated areas can improve their PAO_2 as long as there is inspired O_2 reaches the shunted alveoli. In V/Q mismatch, the shunt is poorly responsive to supplemental O_2 because none of the

V/Q Ratio and Gas Exchange (LO5)

Shunt and dead space, extremes of V/Q mismatch, are typically used to demonstrate the effects of V/Q on alveolar gas exchange. Please note that gas exchange is also affected by less extreme V/Q mismatch.

- In regions with **low $\dot{V}/\dot{Q}$ ratios**, such as with a **shunt** (blocked alveoli), the alveoli are not ventilated and their gases mimic that of the alveolar blood supply to which they are connected. Thus, the alveolar oxygen is reduced and carbon dioxide is elevated (40 and 45 mmHg, respectively), leading to poorly oxygenated blood leaving these areas (**Figure 5**, left image).

- Conversely, lung regions with **high $\dot{V}/\dot{Q}$ ratios**, such as **dead space**, are well ventilated and do not have oxygen removed or carbon dioxide added. Thus, these lung regions show elevated alveolar oxygen and reduced carbon dioxide (150 and 0 mmHg, respectively). But without significant perfusion, these alveoli do not contribute to gas exchange (Figure 5, right image). There is a **decreased effective ventilation**, since exhalation from these regions does not eliminate CO_2 from the blood. Despite this, CO_2 levels typically remain normal because CO_2 's high solubility allows easy elimination through compensatory increases in ventilation, even in severe cases. Hypoxia is common, however. This is because blood is diverted from these areas, resulting in more blood flow to healthy alveoli, over-perfusing them relative to their ventilation and creating low $\dot{V}/\dot{Q}$ regions. These low $\dot{V}/\dot{Q}$ areas lack sufficient ventilation to fully oxygenate the increased blood flow at room air, resulting in hypoxia.

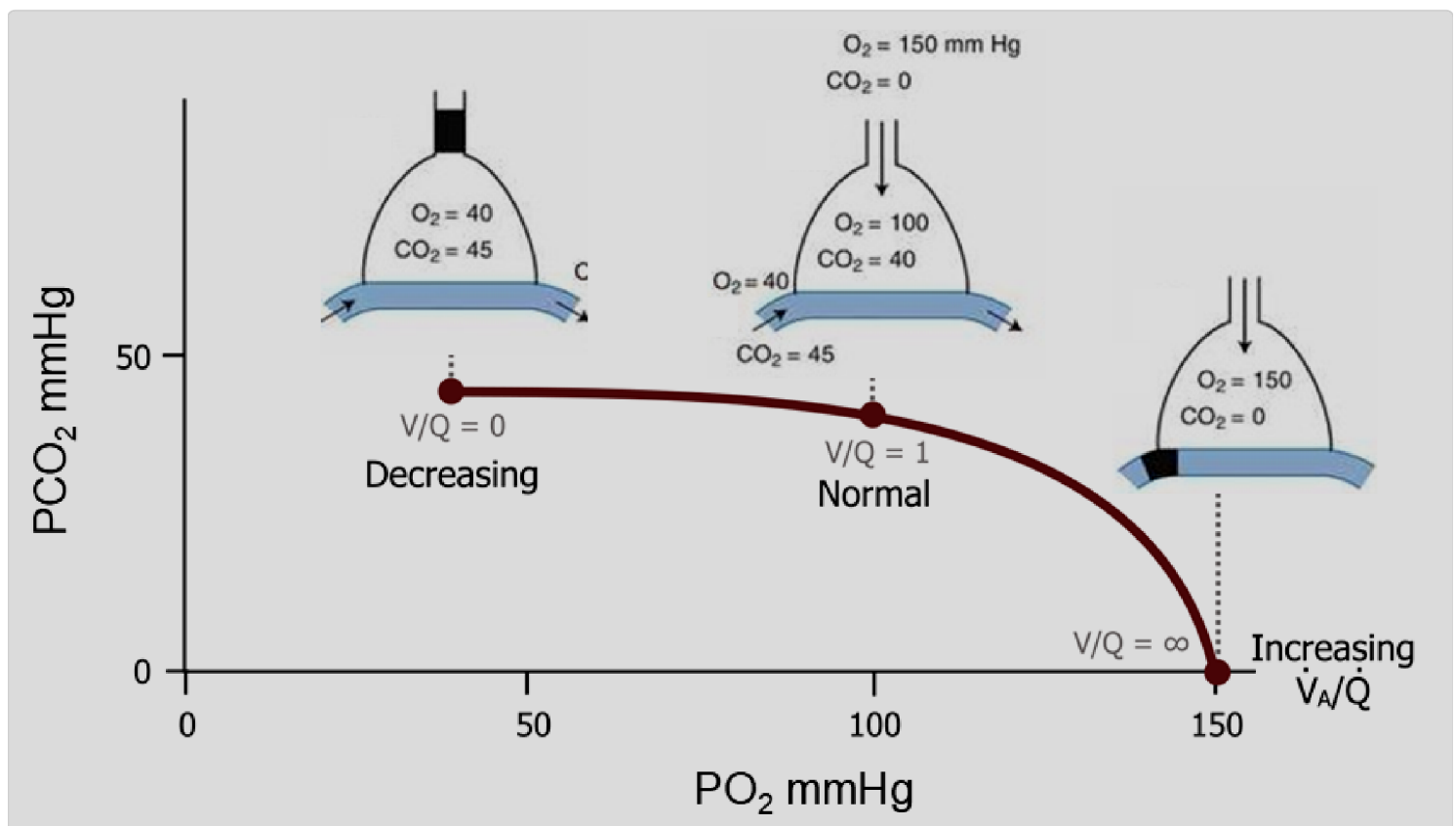


Figure 5. Effect of Dead Space and Shunt, V/Q extremes, on alveolar gases**CLINICAL CORRELATION**

Latent tuberculosis typically reemerges as reactivated tuberculosis at the apex of the lungs where the $\dot{V}/\dot{Q}$ ratio is >1 (hyperventilated region). The excess ventilation at the apex creates an oxygen-rich environment within the alveoli, which allows *Mycobacterium tuberculosis*, an aerobic organism, to thrive.

**CLINICAL REASONING**

If conditions of shunt do not improve with supplemental O₂, how do patients maintain survivable oxygen saturation with conditions like pneumonia and atelectasis?

Regions of the lung that are not ventilated have severely hypoxic alveoli which triggers **hypoxic vasoconstriction** to reduce the shunt fraction and redirect much ($> 50\%$) of the blood flow to healthy alveoli. If hypoxia remains severe, mechanical ventilation with Positive End Expiratory Pressure (PEEP) can recruit collapsed alveoli and decrease shunt, improving oxygenation.

Determining $\dot{V}/\dot{Q}$ States

Clinical assessment of shunt and dead space rely on characteristic patterns of gas exchange abnormalities and response to therapeutic interventions. Shunt

is suggested by severe hypoxemia with poor response to supplemental oxygen, while dead space is indicated by elevated dead space calculations (Bohr equation in the [Alveolar ventilation and dead space](#) brick) and increased ventilatory requirements to maintain normal carbon dioxide levels.

The response to **supplemental oxygen** helps differentiate between different types of $\dot{V}/\dot{Q}$ mismatch. Low $\dot{V}/\dot{Q}$ regions typically show improvement with oxygen therapy because some ventilation is still occurring, allowing the increased alveolar oxygen concentration to improve blood oxygenation. This response helps distinguish low $\dot{V}/\dot{Q}$ states from true shunt conditions.

Advanced [pulmonary function testing](#) can quantify shunt and dead space fractions using specialized equations and measurements. The alveolar-arterial oxygen gradient (A-a gradient, which was covered in the [Alveolar Gas Exchange and Diffusion](#) brick) helps identify the presence of $\dot{V}/\dot{Q}$ mismatch, while the response to 100% oxygen can differentiate between shunt and other causes of hypoxemia. These measurements guide clinical decision-making and monitor response to therapy.

CASE CONNECTION

[BACK TO INTRODUCTION](#) 

CR's oxygen saturation improves with supplemental O_2 , and the CT pulmonary angiogram confirms a pulmonary embolus. Anticoagulation is started.

The embolus obstructs blood flow to part of the lung, creating dead space (ventilated alveoli that receive little or no perfusion), and the blood is diverted to other lung regions, producing V/Q mismatch. His hypoxemia improves with supplemental oxygen because O_2 raises the alveolar PO_2 in the alveoli that are well-ventilated and well-perfused, allowing those healthy regions to compensate for the under-perfused areas. Conversely, conditions that cause shunt (perfused alveoli without ventilation, e.g., airway obstruction, fluid-filled

alveoli, atelectasis) do not respond well to supplemental O_2 , since the oxygen does not reach the alveoli. As a note, hypoxic conditions caused by CO and cyanide poisoning also do not respond well to supplemental oxygen.

Summary

- Pulmonary ventilation involves the movement of air into and out of the alveoli, driven by pressure gradients and respiratory muscle function.
- Perfusion refers to the delivery of deoxygenated blood to alveolar capillaries for oxygenation, influenced by gravity and local factors like alveolar oxygen tension.
- The ventilation/perfusion ($\dot{V}/\dot{Q}$) ratio is crucial for efficient gas exchange, with a normal ratio of approximately 0.8 indicating well-matched ventilation and perfusion.
- In a healthy **upright lung**, $\dot{V}/\dot{Q}$ ratios vary regionally, with higher ratios at the apex and lower ratios at the base due to gravitational effects.
- West zones can be related to different V/Q ratios, higher in the apex, optimal in zone 2, and low in zone 3.
- Shunt occurs when blood is perfused without ventilation, leading to hypoxemia that is poorly responsive to supplemental oxygen.
- A shunt is any region where the lung is perfused but not ventilated, with an extreme shunt having a $\dot{V}/\dot{Q}$ ratio of 0.
- Pathologic examples of a shunt are congenital cardiac right-to-left shunts and complete airway obstruction. Hypoxemia, due to a severe shunt, will **not be corrected with supplemental oxygen**.
- Dead space represents ventilated areas without perfusion, resulting in inefficient CO_2 elimination and requiring increased ventilation to maintain normal CO_2 levels.

- Pathologic examples of dead space include pulmonary embolism and severe heart failure.
- Most lung disease leads to either reduced $\dot{V}/\dot{Q}$ ratio (but not a pure shunt) or increased $\dot{V}/\dot{Q}$ (but not pure dead space).

Review Questions

1. Which of the following scenarios would be most likely to cause a ventilation/perfusion ($\dot{V}/\dot{Q}$) ratio of 0.5?

- A. Destruction of pulmonary capillaries by emphysema
- B. Partial mucus plugging of a bronchiole
- C. Pulmonary embolus
- D. Severe left heart failure
- E. Severe right heart failure

Explanation (requires correct answer)

2. A 65-year-old male presents to the emergency department with pain in his left leg and sudden-onset dyspnea and left-sided chest pain. He is found to have a large pulmonary embolus in his left lung. Which of the following is the most likely $\dot{V}/\dot{Q}$ ratio in the lung region affected by the embolus?

- A. 0
- B. 0.4
- C. 0.8
- D. 1
- E. Infinity

Explanation (requires correct answer)

3. Which of the following correctly compares physiologic values at the base of the lung to those at the apex?

- A. Perfusion and $\dot{V}/\dot{Q}$ ratio are both higher at the base
- B. Perfusion and $\dot{V}/\dot{Q}$ ratio are both lower at the base
- C. Perfusion is higher, and $\dot{V}/\dot{Q}$ ratio is lower at the base
- D. Ventilation and $\dot{V}/\dot{Q}$ ratio are both higher at the base
- E. Ventilation and $\dot{V}/\dot{Q}$ ratio are both lower at the base

Explanation (requires correct answer)

A.

B. Dead space improves with supplemental oxygen, shunt does not

- C. Both improve equally with supplemental oxygen
- D. Neither improves with supplemental oxygen
- E. Both represent a complete absence of blood flow

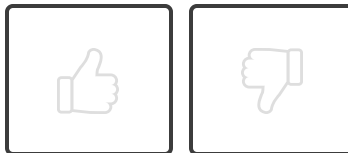
Explanation (requires correct answer)

5. With regard to West zones and perfusion, which of the following best explains why pulmonary perfusion ($\dot{Q}$) is greater at the base of the lung than at the apex in an upright individual?

- A. Higher alveolar oxygen tension at the base increases capillary recruitment
- B. Pulmonary arterial pressure is higher at the base because of gravity
- C. Blood is shunted away from the apex due to hypoxic pulmonary vasoconstriction
- D. Alveolar ventilation is higher at the base than the apex
- E. The pulmonary circulation is a high-pressure system that distributes blood uniformly

Explanation (requires correct answer)

How would you rate this Brick?



Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 Describe the processes of ventilation and perfusion.
- 02 Explain regional differences in pulmonary ventilation and perfusion in the upright healthy lung.
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- 04** Identify the lungs' West zones and the $\dot{V}/\dot{Q}$ ratio of each.
- 05** Compare shunt and dead space in healthy vs pathologic states.

Objective Confidence: 0/5